

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	31.3	10.3	1.5	0.5	0.2	56.2
U3 - Toxicity prior Beta(0.15, 0.85)	33.6	9	0.6	0.4	0.2	56.2
U2 - Toxicity prior Beta(0.3, 0.7)	29.8	12.4	1.6	0.6	0.2	55.4
U3 - Toxicity prior Beta(0.3, 0.7)	33.7	9.7	0.6	0.2	0.1	55.7
U2 - Toxicity prior Beta(0.5, 0.5)	32.7	10.3	2.5	0.6	0.2	53.7
U3 - Toxicity prior Beta(0.5, 0.5)	36	8.8	1.2	0.1	0.2	53.7
U2 - Toxicity prior Beta(1, 1)	33	10.2	2.6	0.6	0.2	53.4
U3 - Toxicity prior Beta(1, 1)	37.8	7.3	1.2	0.1	0.1	53.5
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	15.4	3.9	2.1	0.9	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	15.3	3.9	2.1	0.9	0.4	
U2 - Toxicity prior Beta(0.3, 0.7)	15.2	3.9	2.2	1	0.4	
U3 - Toxicity prior Beta(0.3, 0.7)	15.1	3.9	2.2	1	0.4	
U2 - Toxicity prior Beta(0.5, 0.5)	14.8	4.2	2.4	1.1	0.5	
U3 - Toxicity prior Beta(0.5, 0.5)	14.8	4.2	2.4	1.2	0.5	
U2 - Toxicity prior Beta(1, 1)	14.8	4.2	2.4	1.2	0.5	
U3 - Toxicity prior Beta(1, 1)	14.8	4.2	2.3	1.1	0.5	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.6	2	0.6	0.4	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	2	0.6	0.4	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.6	2	0.6	0.4	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.6	2	0.6	0.4	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.7	1.9	0.8	0.4	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.7	1.9	0.8	0.4	0.1	
U2 - Toxicity prior Beta(1, 1)	0.7	1.9	0.8	0.4	0.1	
U3 - Toxicity prior Beta(1, 1)	0.7	1.9	0.8	0.4	0.1	
Scenario 2						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.2	0.25	0.5	0.5	0.5	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	34.9	16	31.9	11.6	3.8	1.8
U3 - Toxicity prior Beta(0.15, 0.85)	38.3	16.1	30.8	10.1	2.9	1.8
U2 - Toxicity prior Beta(0.3, 0.7)	34.5	14.9	32.2	12	4.6	1.8
U3 - Toxicity prior Beta(0.3, 0.7)	37	16.5	30.8	10.9	3	1.8
U2 - Toxicity prior Beta(0.5, 0.5)	32.7	13.3	32.2	15.1	4.9	1.8
U3 - Toxicity prior Beta(0.5, 0.5)	35.3	15.1	31.3	13.5	3	1.8
U2 - Toxicity prior Beta(1, 1)	31.9	13.9	31.6	15.3	5.5	1.8
U3 - Toxicity prior Beta(1, 1)	35.3	15.3	30.6	13.4	3.6	1.8
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	11.4	5.1	7.2	3.9	2.2	
U3 - Toxicity prior Beta(0.15, 0.85)	11.4	5.1	7.2	3.9	2.2	
U2 - Toxicity prior Beta(0.3, 0.7)	11.2	4.9	7.2	3.9	2.5	
U3 - Toxicity prior Beta(0.3, 0.7)	11.3	4.8	7.2	3.9	2.5	
U2 - Toxicity prior Beta(0.5, 0.5)	10.8	4.6	7.2	4.4	2.7	
U3 - Toxicity prior Beta(0.5, 0.5)	10.8	4.7	7.2	4.4	2.7	
U2 - Toxicity prior Beta(1, 1)	10.9	4.6	7.1	4.5	2.7	
U3 - Toxicity prior Beta(1, 1)	10.8	4.6	7.1	4.5	2.7	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	2.4	1.3	0.9	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	2.4	1.3	0.9	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	2.4	1.3	0.9	0.4	
U3 - Toxicity prior Beta(0.3, 0.7)	0.3	2.4	1.3	0.9	0.4	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	2.3	1.4	1	0.4	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	2.2	1.4	1	0.4	
U2 - Toxicity prior Beta(1, 1)	0.3	2.3	1.4	1	0.4	
U3 - Toxicity prior Beta(1, 1)	0.3	2.2	1.4	1	0.4	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 3						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.1	0.4	0.4	0.4	0.4	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	17.5	40.4	21.1	8.2	2.8	10
U3 - Toxicity prior Beta(0.15, 0.85)	20.2	41.4	20.9	5.4	2.1	10
U2 - Toxicity prior Beta(0.3, 0.7)	17.4	39.1	22	8.3	3.4	9.8
U3 - Toxicity prior Beta(0.3, 0.7)	19.5	41.2	20.5	7	1.9	9.9
U2 - Toxicity prior Beta(0.5, 0.5)	17.1	37	22.9	9.6	3.6	9.8
U3 - Toxicity prior Beta(0.5, 0.5)	19.4	39.5	20.6	8.3	2.4	9.8
U2 - Toxicity prior Beta(1, 1)	16	36.1	24.5	10	3.7	9.7
U3 - Toxicity prior Beta(1, 1)	18.5	40.1	20.7	8.4	2.6	9.7
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9.7	7.5	6.1	3.6	2	
U3 - Toxicity prior Beta(0.15, 0.85)	9.7	7.4	6.2	3.6	2	
U2 - Toxicity prior Beta(0.3, 0.7)	9.6	7.2	6.1	3.7	2.3	
U3 - Toxicity prior Beta(0.3, 0.7)	9.6	7.2	6.2	3.7	2.3	
U2 - Toxicity prior Beta(0.5, 0.5)	9.2	7	6.1	4.1	2.5	
U3 - Toxicity prior Beta(0.5, 0.5)	9.3	6.9	6.2	4.1	2.5	
U2 - Toxicity prior Beta(1, 1)	9.2	6.9	6.2	4.1	2.5	
U3 - Toxicity prior Beta(1, 1)	9.2	6.9	6.2	4.1	2.5	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	2.9	1.3	1	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	2.9	1.3	1	0.4	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.9	1.2	1	0.4	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	2.8	1.3	1.1	0.4	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.7	1.4	1.1	0.4	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.8	1.4	1.1	0.4	
U2 - Toxicity prior Beta(1, 1)	0.2	2.8	1.4	1.1	0.4	
U3 - Toxicity prior Beta(1, 1)	0.2	2.8	1.3	1.1	0.4	
Scenario 4						
DLT rate	0.05	0.1	0.18	0.3	0.4	
Efficacy rate	0.05	0.1	0.25	0.5	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	6.8	7.4	21.4	42.6	19.9	1.9
U3 - Toxicity prior Beta(0.15, 0.85)	11.5	9	19.1	40.8	17.7	1.9
U2 - Toxicity prior Beta(0.3, 0.7)	6.5	7.8	19.5	42	22.3	1.9
U3 - Toxicity prior Beta(0.3, 0.7)	10.8	8.7	17.9	41.8	18.9	1.9
U2 - Toxicity prior Beta(0.5, 0.5)	5.5	6.3	17.8	44.5	24.1	1.8
U3 - Toxicity prior Beta(0.5, 0.5)	9.8	7.4	15.9	44.4	20.7	1.8
U2 - Toxicity prior Beta(1, 1)	5.5	5.6	18.1	45	24	1.8
U3 - Toxicity prior Beta(1, 1)	9	6.9	17.1	44.7	20.5	1.8
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.5	4.2	6.5	8	6.5	
U3 - Toxicity prior Beta(0.15, 0.85)	4.5	4.2	6.5	7.9	6.6	
U2 - Toxicity prior Beta(0.3, 0.7)	4.5	4.1	6.1	8.2	6.9	
U3 - Toxicity prior Beta(0.3, 0.7)	4.4	4.1	6.1	8.1	7	
U2 - Toxicity prior Beta(0.5, 0.5)	4.2	3.8	5.8	8.6	7.3	
U3 - Toxicity prior Beta(0.5, 0.5)	4.2	3.8	5.8	8.7	7.3	
U2 - Toxicity prior Beta(1, 1)	4.2	3.8	5.8	8.7	7.3	
U3 - Toxicity prior Beta(1, 1)	4.2	3.8	5.8	8.6	7.3	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.3	2.8	1.4	2.2	0.8	
U3 - Toxicity prior Beta(0.15, 0.85)	0.3	2.8	1.4	2.2	0.8	
U2 - Toxicity prior Beta(0.3, 0.7)	0.3	2.8	1.4	2.2	0.8	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	2.8	1.4	2.2	0.8	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.6	1.4	2.2	0.8	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.7	1.5	2.2	0.8	
U2 - Toxicity prior Beta(1, 1)	0.2	2.7	1.4	2.2	0.8	
U3 - Toxicity prior Beta(1, 1)	0.2	2.6	1.4	2.2	0.8	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.3	
Efficacy rate	0.05	0.3	0.3	0.3	0.3	
Utility 2	0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	4.9	35.3	26.3	18.1	11.2	4.2
U3 - Toxicity prior Beta(0.15, 0.85)	7.8	41.1	23.4	16.3	7.2	4.2
U2 - Toxicity prior Beta(0.3, 0.7)	4.1	34.7	27.7	17.1	12.2	4.2
U3 - Toxicity prior Beta(0.3, 0.7)	7.9	42.2	22.1	16.3	7.3	4.2
U2 - Toxicity prior Beta(0.5, 0.5)	4.7	33.9	25.1	18.3	13.8	4.2
U3 - Toxicity prior Beta(0.5, 0.5)	7.8	40.1	22.9	16.3	8.7	4.2
U2 - Toxicity prior Beta(1, 1)	4.3	34.3	25.7	19.7	11.8	4.2
U3 - Toxicity prior Beta(1, 1)	7.9	39.9	24.5	15.4	8.1	4.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	5	6.4	6.7	5.7	5.7	
U3 - Toxicity prior Beta(0.15, 0.85)	5	6.4	6.7	5.7	5.6	
U2 - Toxicity prior Beta(0.3, 0.7)	4.9	6.3	6.7	5.5	6	
U3 - Toxicity prior Beta(0.3, 0.7)	5	6.4	6.5	5.6	5.9	
U2 - Toxicity prior Beta(0.5, 0.5)	4.9	6.2	6.5	5.6	6.2	
U3 - Toxicity prior Beta(0.5, 0.5)	4.9	6.1	6.5	5.7	6.2	
U2 - Toxicity prior Beta(1, 1)	4.9	6	6.5	5.7	6.2	
U3 - Toxicity prior Beta(1, 1)	4.9	6	6.5	5.8	6.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	3.4	1.8	2.1	1.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	3.4	1.7	2.1	1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	3.4	1.8	2.1	1.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	3.4	1.7	2.1	1.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	3.3	1.8	2	1.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	3.3	1.9	2	1.1	
U2 - Toxicity prior Beta(1, 1)	0.2	3.3	1.9	2	1.1	
U3 - Toxicity prior Beta(1, 1)	0.2	3.3	1.8	2.1	1.1	
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.3	0.32	0.35	0.38	0.4	
Utility 2	0	0	0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	67.6	14.2	3.2	1.2	0.5	13.3
U3 - Toxicity prior Beta(0.15, 0.85)	68.7	13.3	3.1	1	0.4	13.5
U2 - Toxicity prior Beta(0.3, 0.7)	66.9	13.9	3.8	1.4	0.5	13.5
U3 - Toxicity prior Beta(0.3, 0.7)	68.7	13.3	3.1	0.9	0.4	13.6
U2 - Toxicity prior Beta(0.5, 0.5)	65.9	13.1	5.4	1.6	0.6	13.4
U3 - Toxicity prior Beta(0.5, 0.5)	67.8	13.3	4.5	1.1	0.3	13
U2 - Toxicity prior Beta(1, 1)	64.8	13.8	6	1.9	0.5	13
U3 - Toxicity prior Beta(1, 1)	66.6	13.5	4.8	1.6	0.3	13.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	19.6	3.9	2.5	1.1	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	19.6	3.9	2.5	1.1	0.4	
U2 - Toxicity prior Beta(0.3, 0.7)	19.4	3.8	2.6	1.2	0.5	
U3 - Toxicity prior Beta(0.3, 0.7)	19.5	3.8	2.6	1.2	0.5	
U2 - Toxicity prior Beta(0.5, 0.5)	18.8	4.1	2.7	1.4	0.6	
U3 - Toxicity prior Beta(0.5, 0.5)	18.8	4.1	2.7	1.4	0.6	
U2 - Toxicity prior Beta(1, 1)	18.8	4.1	2.7	1.4	0.6	
U3 - Toxicity prior Beta(1, 1)	18.8	4	2.7	1.5	0.6	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	1.5	0.5	0.3	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	1.5	0.5	0.3	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	1.5	0.6	0.3	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	1.5	0.6	0.3	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	1.5	0.6	0.4	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	1.5	0.6	0.4	0.1	
U2 - Toxicity prior Beta(1, 1)	0.4	1.4	0.6	0.4	0.1	
U3 - Toxicity prior Beta(1, 1)	0.4	1.4	0.6	0.4	0.1	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 7						
DLT rate	0.16	0.2	0.26	0.31	0.36	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	26.4	36.6	20.5	10.6	3.7	2.2
U3 - Toxicity prior Beta(0.15, 0.85)	28.2	39	18.9	8.6	3	2.3
U2 - Toxicity prior Beta(0.3, 0.7)	26.1	35.5	21.4	10	4.8	2.2
U3 - Toxicity prior Beta(0.3, 0.7)	27.9	37.9	19.6	9	3.4	2.2
U2 - Toxicity prior Beta(0.5, 0.5)	25.6	32.4	21.9	11.9	6	2.2
U3 - Toxicity prior Beta(0.5, 0.5)	27	35.9	21.9	9.4	3.6	2.2
U2 - Toxicity prior Beta(1, 1)	25.6	32.9	22.3	11.5	5.5	2.2
U3 - Toxicity prior Beta(1, 1)	27.4	36.4	20.2	10.1	3.7	2.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	10.8	7	5.6	3.8	2.5	
U3 - Toxicity prior Beta(0.15, 0.85)	10.8	6.9	5.7	3.7	2.6	
U2 - Toxicity prior Beta(0.3, 0.7)	10.7	6.7	5.6	3.8	2.8	
U3 - Toxicity prior Beta(0.3, 0.7)	10.7	6.8	5.6	3.8	2.8	
U2 - Toxicity prior Beta(0.5, 0.5)	10.3	6.4	5.8	4	3.1	
U3 - Toxicity prior Beta(0.5, 0.5)	10.3	6.5	5.8	4	3.1	
U2 - Toxicity prior Beta(1, 1)	10.3	6.4	5.7	4.1	3.1	
U3 - Toxicity prior Beta(1, 1)	10.3	6.5	5.7	4.1	3	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	2.6	1.1	0.9	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	2.6	1.1	0.9	0.4	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.6	1.1	1	0.4	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	2.6	1.1	1	0.4	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.5	1.2	1	0.4	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.5	1.2	1	0.4	
U2 - Toxicity prior Beta(1, 1)	0.2	2.5	1.2	1	0.4	
U3 - Toxicity prior Beta(1, 1)	0.2	2.5	1.1	1	0.4	
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.1	0.15	0.2	0.3	0.4	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	43.2	9.2	2.8	1.2	0.3	43.3
U3 - Toxicity prior Beta(0.15, 0.85)	45	8.7	2	0.9	0.2	43.2
U2 - Toxicity prior Beta(0.3, 0.7)	42.1	9.9	3.5	0.9	0.4	43.2
U3 - Toxicity prior Beta(0.3, 0.7)	43.5	9.6	2.2	0.9	0.4	43.4
U2 - Toxicity prior Beta(0.5, 0.5)	41.1	10.1	4.4	1.3	0.8	42.3
U3 - Toxicity prior Beta(0.5, 0.5)	43.1	9.9	3.2	1.2	0.4	42.2
U2 - Toxicity prior Beta(1, 1)	41.9	9.4	3.9	1.9	0.8	42.1
U3 - Toxicity prior Beta(1, 1)	43.1	10.3	3.1	1.1	0.5	41.9
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	17.3	3.7	1.9	0.8	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	17.4	3.7	1.9	0.8	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	17.1	3.7	2	0.9	0.4	
U3 - Toxicity prior Beta(0.3, 0.7)	17.2	3.7	1.9	0.9	0.4	
U2 - Toxicity prior Beta(0.5, 0.5)	16.6	4.1	2.1	1	0.4	
U3 - Toxicity prior Beta(0.5, 0.5)	16.6	4.1	2.1	1	0.4	
U2 - Toxicity prior Beta(1, 1)	16.6	4	2.1	1.1	0.4	
U3 - Toxicity prior Beta(1, 1)	16.6	4	2.1	1	0.4	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.3	1.6	0.4	0.3	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.3	1.6	0.4	0.2	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	1.6	0.5	0.3	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	1.6	0.4	0.3	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	1.5	0.6	0.3	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	1.5	0.6	0.3	0.1	
U2 - Toxicity prior Beta(1, 1)	0.4	1.5	0.6	0.3	0.1	
U3 - Toxicity prior Beta(1, 1)	0.4	1.5	0.6	0.3	0.1	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.2	0.25	0.5	
Utility 2	-0.1	-0.1	0	0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	13.4	10.5	17.5	13.2	26.8	18.6
U3 - Toxicity prior Beta(0.15, 0.85)	20.9	10.6	14.8	12	23	18.7
U2 - Toxicity prior Beta(0.3, 0.7)	12.8	9	17.2	13.5	28.5	19
U3 - Toxicity prior Beta(0.3, 0.7)	18.8	11	14.4	12.4	24.6	18.8
U2 - Toxicity prior Beta(0.5, 0.5)	12.7	8.4	16.4	13.1	30.9	18.5
U3 - Toxicity prior Beta(0.5, 0.5)	16.6	10	15.5	12.7	26.9	18.3
U2 - Toxicity prior Beta(1, 1)	11.8	8.4	16.9	13.5	31.2	18.2
U3 - Toxicity prior Beta(1, 1)	17.7	9.9	14.8	12.3	27	18.3
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	8.9	4.3	5.3	4	5.3	
U3 - Toxicity prior Beta(0.15, 0.85)	8.9	4.3	5.3	4	5.3	
U2 - Toxicity prior Beta(0.3, 0.7)	8.7	4.2	5.2	4	5.7	
U3 - Toxicity prior Beta(0.3, 0.7)	8.7	4.2	5.1	4	5.8	
U2 - Toxicity prior Beta(0.5, 0.5)	8.4	3.9	5	4.3	6.3	
U3 - Toxicity prior Beta(0.5, 0.5)	8.3	4	5	4.4	6.2	
U2 - Toxicity prior Beta(1, 1)	8.4	3.9	5	4.4	6.2	
U3 - Toxicity prior Beta(1, 1)	8.4	3.9	5	4.3	6.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	2.6	1.3	1.4	0.7	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	2.6	1.3	1.4	0.7	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	2.6	1.3	1.5	0.7	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	2.6	1.3	1.4	0.7	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	2.4	1.4	1.4	0.8	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	2.4	1.4	1.5	0.7	
U2 - Toxicity prior Beta(1, 1)	0.4	2.4	1.4	1.4	0.8	
U3 - Toxicity prior Beta(1, 1)	0.4	2.4	1.4	1.4	0.8	
Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.4	0.42	0.45	0.5	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	11.4	31.7	11.8	4.6	2.4	38.1
U3 - Toxicity prior Beta(0.15, 0.85)	12.4	32.5	10.7	4.6	1.5	38.3
U2 - Toxicity prior Beta(0.3, 0.7)	11.1	30.2	13.2	4.9	2.8	37.8
U3 - Toxicity prior Beta(0.3, 0.7)	11.9	33.1	11	4.4	2.1	37.5
U2 - Toxicity prior Beta(0.5, 0.5)	10.4	28.5	14.6	6.5	3.4	36.6
U3 - Toxicity prior Beta(0.5, 0.5)	11.5	30.3	13.2	6	2.3	36.7
U2 - Toxicity prior Beta(1, 1)	10.7	27.8	14.3	6.8	3.6	36.8
U3 - Toxicity prior Beta(1, 1)	12	29.8	13.3	5.8	2.4	36.7
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	11.7	5.8	4.1	2.2	1.2	
U3 - Toxicity prior Beta(0.15, 0.85)	11.7	5.9	4	2.2	1.2	
U2 - Toxicity prior Beta(0.3, 0.7)	11.6	5.6	4.3	2.2	1.4	
U3 - Toxicity prior Beta(0.3, 0.7)	11.6	5.7	4.1	2.3	1.4	
U2 - Toxicity prior Beta(0.5, 0.5)	10.9	6	4.3	2.7	1.5	
U3 - Toxicity prior Beta(0.5, 0.5)	11	5.9	4.2	2.7	1.6	
U2 - Toxicity prior Beta(1, 1)	11	6	4.2	2.7	1.6	
U3 - Toxicity prior Beta(1, 1)	11	5.9	4.2	2.7	1.6	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.1	2.2	0.8	0.6	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.1	2.2	0.8	0.6	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	2.2	0.9	0.6	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.1	2.2	0.8	0.6	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2.1	1	0.7	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	2.1	1	0.7	0.2	
U2 - Toxicity prior Beta(1, 1)	0.1	2.1	1	0.7	0.2	
U3 - Toxicity prior Beta(1, 1)	0.1	2.1	1	0.6	0.2	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	65.9	22	4.6	0.5	0	7
U3 - Toxicity prior Beta(0.15, 0.85)	69.5	20.6	2.4	0.7	0	6.8
U2 - Toxicity prior Beta(0.3, 0.7)	65.9	21.9	4.6	0.8	0.1	6.7
U3 - Toxicity prior Beta(0.3, 0.7)	69.2	20.9	2.7	0.5	0	6.7
U2 - Toxicity prior Beta(0.5, 0.5)	65.8	21.3	5.5	0.9	0.1	6.4
U3 - Toxicity prior Beta(0.5, 0.5)	70.2	18.8	3.5	0.9	0	6.6
U2 - Toxicity prior Beta(1, 1)	66.3	20.5	5.9	0.8	0.1	6.4
U3 - Toxicity prior Beta(1, 1)	69.2	20.3	3.4	0.8	0	6.3
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	18.4	5.8	3.5	1.1	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	18.4	5.8	3.5	1.1	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	18.2	5.7	3.6	1.2	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	18.2	5.7	3.6	1.2	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	17.5	6	3.6	1.6	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	17.6	5.9	3.7	1.5	0.3	
U2 - Toxicity prior Beta(1, 1)	17.6	5.9	3.7	1.5	0.3	
U3 - Toxicity prior Beta(1, 1)	17.5	6	3.7	1.5	0.3	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.6	2.4	0.8	0.4	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.6	2.4	0.8	0.4	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.6	2.3	0.8	0.4	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.6	2.4	0.8	0.4	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.7	2.2	0.9	0.5	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.7	2.2	1	0.5	0.1	
U2 - Toxicity prior Beta(1, 1)	0.7	2.2	1	0.5	0.1	
U3 - Toxicity prior Beta(1, 1)	0.7	2.2	1	0.5	0.1	
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.1	0.12	0.4	0.6	
Utility 2	0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	14.2	12.3	10	36	24.2	3.3
U3 - Toxicity prior Beta(0.15, 0.85)	21.2	12.2	6.8	34.4	22.1	3.3
U2 - Toxicity prior Beta(0.3, 0.7)	14.4	11.5	9	37	24.8	3.3
U3 - Toxicity prior Beta(0.3, 0.7)	18.7	11.9	6.6	36	23.4	3.4
U2 - Toxicity prior Beta(0.5, 0.5)	12	9.6	7.7	38	29.5	3.2
U3 - Toxicity prior Beta(0.5, 0.5)	16.8	9.8	6.5	37.2	26.5	3.2
U2 - Toxicity prior Beta(1, 1)	11.5	9.2	7.4	37.8	30.9	3.2
U3 - Toxicity prior Beta(1, 1)	17.6	9.8	5.8	35.9	27.7	3.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	5	4.5	5.3	7.3	7.4	
U3 - Toxicity prior Beta(0.15, 0.85)	5	4.6	5.2	7.3	7.3	
U2 - Toxicity prior Beta(0.3, 0.7)	4.9	4.4	4.9	7.5	7.8	
U3 - Toxicity prior Beta(0.3, 0.7)	5	4.3	4.9	7.6	7.8	
U2 - Toxicity prior Beta(0.5, 0.5)	4.7	4.1	4.6	7.9	8.2	
U3 - Toxicity prior Beta(0.5, 0.5)	4.6	4.1	4.7	8	8.2	
U2 - Toxicity prior Beta(1, 1)	4.7	4.1	4.6	8	8.2	
U3 - Toxicity prior Beta(1, 1)	4.7	4.1	4.6	7.9	8.3	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.3	2.9	1.4	2.2	0.8	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	2.9	1.4	2.2	0.8	
U2 - Toxicity prior Beta(0.3, 0.7)	0.3	2.8	1.3	2.2	0.8	
U3 - Toxicity prior Beta(0.3, 0.7)	0.3	2.8	1.3	2.2	0.8	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	2.7	1.4	2.1	0.9	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	2.7	1.4	2.1	0.9	
U2 - Toxicity prior Beta(1, 1)	0.3	2.7	1.3	2.1	0.9	
U3 - Toxicity prior Beta(1, 1)	0.3	2.7	1.3	2.1	0.8	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 13						
DLT rate	0.21	0.29	0.3	0.39	0.46	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.2	0	0	0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	13.8	32.8	13.4	4.6	2	33.4
U3 - Toxicity prior Beta(0.15, 0.85)	15.8	32.7	12.6	4.4	1.6	32.9
U2 - Toxicity prior Beta(0.3, 0.7)	13.2	31.4	15.6	5.2	1.9	32.7
U3 - Toxicity prior Beta(0.3, 0.7)	14.4	33.2	13.1	5.1	1.6	32.6
U2 - Toxicity prior Beta(0.5, 0.5)	12.8	30.2	16.1	6.7	2.6	31.6
U3 - Toxicity prior Beta(0.5, 0.5)	14.4	31.4	15.1	5.6	1.9	31.6
U2 - Toxicity prior Beta(1, 1)	12.2	30.8	16	6.9	2.5	31.6
U3 - Toxicity prior Beta(1, 1)	14.2	31.6	14.7	5.8	2	31.7
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	11.4	6.1	4.5	2.5	1.2	
U3 - Toxicity prior Beta(0.15, 0.85)	11.4	6.1	4.5	2.4	1.2	
U2 - Toxicity prior Beta(0.3, 0.7)	11.3	5.9	4.7	2.5	1.4	
U3 - Toxicity prior Beta(0.3, 0.7)	11.2	6	4.6	2.5	1.4	
U2 - Toxicity prior Beta(0.5, 0.5)	10.5	6.3	4.8	2.8	1.6	
U3 - Toxicity prior Beta(0.5, 0.5)	10.5	6.4	4.7	2.8	1.6	
U2 - Toxicity prior Beta(1, 1)	10.5	6.3	4.7	2.9	1.6	
U3 - Toxicity prior Beta(1, 1)	10.5	6.3	4.7	2.9	1.6	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	2.5	0.9	0.8	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	2.5	0.9	0.7	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.5	1	0.8	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	2.5	0.9	0.8	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.4	1.1	0.8	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.4	1.1	0.8	0.2	
U2 - Toxicity prior Beta(1, 1)	0.2	2.3	1.1	0.8	0.2	
U3 - Toxicity prior Beta(1, 1)	0.2	2.3	1.1	0.8	0.2	
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.3	0.4	
Utility 2	0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	19.6	14.5	12.8	27.8	18	7.3
U3 - Toxicity prior Beta(0.15, 0.85)	30.5	15.1	9	25.3	12.7	7.4
U2 - Toxicity prior Beta(0.3, 0.7)	19.1	13.9	12	29.6	18.2	7.2
U3 - Toxicity prior Beta(0.3, 0.7)	29.7	14.7	9.2	24.9	14.5	7
U2 - Toxicity prior Beta(0.5, 0.5)	17.7	12.8	10.2	31.5	20.7	7.1
U3 - Toxicity prior Beta(0.5, 0.5)	24.7	14.6	8.4	29.1	16.1	7.1
U2 - Toxicity prior Beta(1, 1)	16.8	12.6	11.2	30.8	21.5	7.1
U3 - Toxicity prior Beta(1, 1)	25.7	14.9	7.8	27.8	16.8	7
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	6.7	5.2	5.8	6.1	5.3	
U3 - Toxicity prior Beta(0.15, 0.85)	6.7	5.2	5.8	6.1	5.2	
U2 - Toxicity prior Beta(0.3, 0.7)	6.6	5.1	5.5	6.4	5.6	
U3 - Toxicity prior Beta(0.3, 0.7)	6.5	5	5.5	6.4	5.6	
U2 - Toxicity prior Beta(0.5, 0.5)	6.3	4.7	5.3	6.9	6	
U3 - Toxicity prior Beta(0.5, 0.5)	6.2	4.6	5.3	6.9	6	
U2 - Toxicity prior Beta(1, 1)	6.2	4.7	5.3	6.9	6	
U3 - Toxicity prior Beta(1, 1)	6.3	4.7	5.3	6.9	6	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.6	3.1	1.5	2.1	0.8	
U3 - Toxicity prior Beta(0.15, 0.85)	0.6	3.1	1.5	2	0.8	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	3	1.4	2.1	0.8	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	3	1.4	2.1	0.8	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	2.9	1.5	2.1	0.9	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	2.9	1.5	2.1	0.9	
U2 - Toxicity prior Beta(1, 1)	0.4	2.9	1.5	2.1	0.9	
U3 - Toxicity prior Beta(1, 1)	0.5	2.9	1.5	2.1	0.9	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 15						
DLT rate	0.13	0.2	0.32	0.39	0.46	
Efficacy rate	0.1	0.1	0.1	0.1	0.1	
Utility 2	0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	52.4	24.9	10.9	3.6	1.2	7
U3 - Toxicity prior Beta(0.15, 0.85)	63.8	20.5	5.7	2.4	0.4	7.2
U2 - Toxicity prior Beta(0.3, 0.7)	52.9	25	10.1	4	1.1	6.9
U3 - Toxicity prior Beta(0.3, 0.7)	64.5	20.2	5.8	2.2	0.3	7
U2 - Toxicity prior Beta(0.5, 0.5)	52.2	23.6	11.6	4.4	1	7.2
U3 - Toxicity prior Beta(0.5, 0.5)	63.9	20.9	6.1	1.9	0.1	7.1
U2 - Toxicity prior Beta(1, 1)	53.6	22.3	11	5.1	0.8	7.2
U3 - Toxicity prior Beta(1, 1)	65.4	19.3	5.7	2.2	0.2	7.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	11.6	6.6	5.9	3.3	1.7	
U3 - Toxicity prior Beta(0.15, 0.85)	11.7	6.6	5.8	3.3	1.7	
U2 - Toxicity prior Beta(0.3, 0.7)	11.5	6.5	5.8	3.4	1.9	
U3 - Toxicity prior Beta(0.3, 0.7)	11.4	6.5	5.8	3.5	1.9	
U2 - Toxicity prior Beta(0.5, 0.5)	11.1	6.3	5.8	4	2.1	
U3 - Toxicity prior Beta(0.5, 0.5)	11.1	6.3	5.7	4	2.1	
U2 - Toxicity prior Beta(1, 1)	11	6.2	5.7	4	2.2	
U3 - Toxicity prior Beta(1, 1)	11.1	6.2	5.8	4	2.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	1.1	3.4	1.6	1.3	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	1.1	3.4	1.6	1.3	0.4	
U2 - Toxicity prior Beta(0.3, 0.7)	1	3.3	1.6	1.3	0.4	
U3 - Toxicity prior Beta(0.3, 0.7)	1	3.3	1.6	1.3	0.4	
U2 - Toxicity prior Beta(0.5, 0.5)	1	3.2	1.8	1.4	0.5	
U3 - Toxicity prior Beta(0.5, 0.5)	1	3.2	1.8	1.4	0.5	
U2 - Toxicity prior Beta(1, 1)	1	3.1	1.8	1.4	0.5	
U3 - Toxicity prior Beta(1, 1)	1	3.1	1.7	1.4	0.5	
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	31.4	23.2	16.2	16.6	12.5	0.1
U3 - Toxicity prior Beta(0.15, 0.85)	38.2	25.7	15.6	12.9	7.5	0.1
U2 - Toxicity prior Beta(0.3, 0.7)	31.2	23.5	16.2	15.9	13.1	0.1
U3 - Toxicity prior Beta(0.3, 0.7)	37.6	25.7	15.3	13.2	8.1	0.1
U2 - Toxicity prior Beta(0.5, 0.5)	29.8	23.2	16.5	16.2	14.2	0.1
U3 - Toxicity prior Beta(0.5, 0.5)	37.4	25.6	14.8	13.5	8.6	0.1
U2 - Toxicity prior Beta(1, 1)	31.1	22.1	15.8	16.1	14.8	0.1
U3 - Toxicity prior Beta(1, 1)	36	27.5	15	13	8.4	0.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	6.1	5.4	5.4	5.8	7.3	
U3 - Toxicity prior Beta(0.15, 0.85)	6	5.4	5.5	5.8	7.3	
U2 - Toxicity prior Beta(0.3, 0.7)	6.1	5.4	5.4	5.7	7.4	
U3 - Toxicity prior Beta(0.3, 0.7)	6	5.4	5.5	5.7	7.4	
U2 - Toxicity prior Beta(0.5, 0.5)	5.9	5.4	5.4	5.7	7.6	
U3 - Toxicity prior Beta(0.5, 0.5)	6	5.3	5.4	5.6	7.6	
U2 - Toxicity prior Beta(1, 1)	5.9	5.3	5.4	5.7	7.6	
U3 - Toxicity prior Beta(1, 1)	6	5.3	5.5	5.7	7.6	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	1.1	3.1	1.8	2.6	1.5	
U3 - Toxicity prior Beta(0.15, 0.85)	1.1	3.1	1.8	2.6	1.5	
U2 - Toxicity prior Beta(0.3, 0.7)	1.1	3.1	1.8	2.6	1.4	
U3 - Toxicity prior Beta(0.3, 0.7)	1.1	3.1	1.9	2.6	1.4	
U2 - Toxicity prior Beta(0.5, 0.5)	1	3.1	1.9	2.6	1.4	
U3 - Toxicity prior Beta(0.5, 0.5)	1.1	3.1	1.9	2.6	1.4	
U2 - Toxicity prior Beta(1, 1)	1.1	3.1	1.9	2.6	1.4	
U3 - Toxicity prior Beta(1, 1)	1.1	3.1	1.9	2.6	1.4	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.3	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	30.3	49.6	17.9	1.9	0	0.3
U3 - Toxicity prior Beta(0.15, 0.85)	34	49.1	14.8	1.8	0	0.3
U2 - Toxicity prior Beta(0.3, 0.7)	30.8	48.1	18.7	2.1	0	0.3
U3 - Toxicity prior Beta(0.3, 0.7)	34.7	47.2	15.9	1.9	0	0.3
U2 - Toxicity prior Beta(0.5, 0.5)	30.5	46.8	19.8	2.6	0	0.3
U3 - Toxicity prior Beta(0.5, 0.5)	33.8	47.4	16.3	2.2	0	0.3
U2 - Toxicity prior Beta(1, 1)	29.8	46.7	20	3.2	0	0.3
U3 - Toxicity prior Beta(1, 1)	33.2	47	16.9	2.6	0	0.3
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	10.2	9.7	7.3	2.4	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	10.3	9.7	7.2	2.3	0.5	
U2 - Toxicity prior Beta(0.3, 0.7)	10.1	9.3	7.4	2.6	0.5	
U3 - Toxicity prior Beta(0.3, 0.7)	10.1	9.4	7.3	2.6	0.5	
U2 - Toxicity prior Beta(0.5, 0.5)	9.5	9.1	7.5	3.2	0.6	
U3 - Toxicity prior Beta(0.5, 0.5)	9.6	9.1	7.5	3.2	0.6	
U2 - Toxicity prior Beta(1, 1)	9.6	9	7.5	3.2	0.6	
U3 - Toxicity prior Beta(1, 1)	9.6	9	7.6	3.1	0.6	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	2.8	1.1	0.6	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	2.8	1.1	0.6	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	2.8	1.1	0.6	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	2.8	1.1	0.6	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	2.7	1.2	0.6	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	2.7	1.2	0.7	0.1	
U2 - Toxicity prior Beta(1, 1)	0.4	2.7	1.2	0.7	0.1	
U3 - Toxicity prior Beta(1, 1)	0.4	2.7	1.2	0.7	0.1	
Scenario 18						
DLT rate	0.02	0.04	0.1	0.2	0.35	
Efficacy rate	0.05	0.1	0.15	0.2	0.5	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	9.8	7.5	9.2	14.2	58.5	0.8
U3 - Toxicity prior Beta(0.15, 0.85)	16.1	10	9.6	11.8	51.7	0.8
U2 - Toxicity prior Beta(0.3, 0.7)	9.5	8	8.2	14	59.5	0.8
U3 - Toxicity prior Beta(0.3, 0.7)	15.7	9.2	8.8	12.1	53.4	0.8
U2 - Toxicity prior Beta(0.5, 0.5)	8.7	7	7.9	13.2	62.5	0.7
U3 - Toxicity prior Beta(0.5, 0.5)	14.2	9.9	8.9	11.3	55	0.7
U2 - Toxicity prior Beta(1, 1)	7.7	7.6	6.6	12.7	64.7	0.7
U3 - Toxicity prior Beta(1, 1)	13.5	9.8	8.9	11	56.1	0.7
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.8	3.8	4.6	5.5	12.2	
U3 - Toxicity prior Beta(0.15, 0.85)	3.8	3.9	4.6	5.5	12.2	
U2 - Toxicity prior Beta(0.3, 0.7)	3.8	3.8	4.4	5.4	12.6	
U3 - Toxicity prior Beta(0.3, 0.7)	3.7	3.8	4.4	5.4	12.6	
U2 - Toxicity prior Beta(0.5, 0.5)	3.7	3.8	4.3	5.4	12.8	
U3 - Toxicity prior Beta(0.5, 0.5)	3.7	3.8	4.3	5.4	12.7	
U2 - Toxicity prior Beta(1, 1)	3.7	3.8	4.3	5.4	12.7	
U3 - Toxicity prior Beta(1, 1)	3.7	3.7	4.3	5.4	12.8	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	2.7	1.4	2.3	1.4	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	2.8	1.4	2.3	1.4	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.7	1.4	2.3	1.4	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	2.8	1.4	2.3	1.4	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.7	1.4	2.3	1.4	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.7	1.4	2.3	1.4	
U2 - Toxicity prior Beta(1, 1)	0.2	2.7	1.4	2.3	1.4	
U3 - Toxicity prior Beta(1, 1)	0.2	2.7	1.4	2.3	1.4	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.6	0.77	
Efficacy rate	0.08	0.3	0.32	0.35	0.4	
Utility 2	0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	20.6	51.9	17.5	1.6	0	8.4
U3 - Toxicity prior Beta(0.15, 0.85)	23.3	52.2	15.1	1.2	0	8.2
U2 - Toxicity prior Beta(0.3, 0.7)	19.9	51.3	18.6	1.9	0.1	8.2
U3 - Toxicity prior Beta(0.3, 0.7)	23	51.4	15.8	1.4	0.1	8.3
U2 - Toxicity prior Beta(0.5, 0.5)	18.5	49.7	21.6	1.9	0.2	8.1
U3 - Toxicity prior Beta(0.5, 0.5)	23.4	49	18	1.6	0.1	7.9
U2 - Toxicity prior Beta(1, 1)	18	50.1	21.6	2	0.3	8
U3 - Toxicity prior Beta(1, 1)	22.1	49.8	18	1.9	0.2	8
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9.2	10.2	7	2.2	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	9.1	10.2	7.1	2.3	0.4	
U2 - Toxicity prior Beta(0.3, 0.7)	9	9.9	7.2	2.5	0.4	
U3 - Toxicity prior Beta(0.3, 0.7)	8.9	9.8	7.3	2.5	0.5	
U2 - Toxicity prior Beta(0.5, 0.5)	8.3	9.7	7.5	3.1	0.5	
U3 - Toxicity prior Beta(0.5, 0.5)	8.4	9.8	7.3	3.1	0.5	
U2 - Toxicity prior Beta(1, 1)	8.3	9.8	7.4	3	0.5	
U3 - Toxicity prior Beta(1, 1)	8.4	9.7	7.4	3.1	0.5	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	3.6	1.3	0.7	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	3.5	1.3	0.7	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	3.5	1.3	0.8	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	3.5	1.3	0.8	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	3.3	1.5	0.8	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	3.2	1.5	0.8	0.1	
U2 - Toxicity prior Beta(1, 1)	0.4	3.2	1.5	0.8	0.1	
U3 - Toxicity prior Beta(1, 1)	0.4	3.2	1.5	0.8	0.1	
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	12.3	44.8	27.7	9.7	1.2	4.3
U3 - Toxicity prior Beta(0.15, 0.85)	18.1	45.8	23.4	7.8	0.6	4.3
U2 - Toxicity prior Beta(0.3, 0.7)	12	43.1	28.4	10.5	1.7	4.3
U3 - Toxicity prior Beta(0.3, 0.7)	17.9	45.4	23.5	8	0.9	4.3
U2 - Toxicity prior Beta(0.5, 0.5)	11.8	41.3	28	12	2.7	4.2
U3 - Toxicity prior Beta(0.5, 0.5)	18.5	45.1	20.8	9.8	1.6	4.2
U2 - Toxicity prior Beta(1, 1)	12.6	40.6	27.5	13	2.1	4.2
U3 - Toxicity prior Beta(1, 1)	18.8	45.1	21.2	9.1	1.6	4.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	5.6	8.7	8.3	4.6	2.2	
U3 - Toxicity prior Beta(0.15, 0.85)	5.7	8.5	8.3	4.6	2.3	
U2 - Toxicity prior Beta(0.3, 0.7)	5.6	8.4	8.1	4.9	2.5	
U3 - Toxicity prior Beta(0.3, 0.7)	5.6	8.2	8.2	5	2.4	
U2 - Toxicity prior Beta(0.5, 0.5)	5.4	7.8	7.8	5.7	2.7	
U3 - Toxicity prior Beta(0.5, 0.5)	5.4	7.8	7.8	5.7	2.7	
U2 - Toxicity prior Beta(1, 1)	5.4	7.6	7.9	5.8	2.7	
U3 - Toxicity prior Beta(1, 1)	5.4	7.7	7.9	5.7	2.7	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	3.7	1.8	1.6	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	3.6	1.8	1.6	0.5	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	3.6	1.8	1.6	0.5	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	3.6	1.8	1.6	0.5	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	3.4	1.9	1.7	0.6	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	3.4	1.9	1.7	0.6	
U2 - Toxicity prior Beta(1, 1)	0.4	3.4	1.9	1.7	0.6	
U3 - Toxicity prior Beta(1, 1)	0.4	3.4	1.9	1.7	0.5	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.5	0.5	0.5	0.5	0.5	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	59.9	1.6	0	0	0	38.5
U3 - Toxicity prior Beta(0.15, 0.85)	60.5	1.4	0	0	0	38.1
U2 - Toxicity prior Beta(0.3, 0.7)	60.3	1.8	0	0	0	37.9
U3 - Toxicity prior Beta(0.3, 0.7)	61	1.2	0	0	0	37.8
U2 - Toxicity prior Beta(0.5, 0.5)	60.9	1.7	0	0	0	37.4
U3 - Toxicity prior Beta(0.5, 0.5)	61.2	1.4	0	0	0	37.4
U2 - Toxicity prior Beta(1, 1)	60.6	1.7	0	0	0	37.7
U3 - Toxicity prior Beta(1, 1)	60.8	1.4	0	0	0	37.8
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	20.9	2.1	0.6	0	0	
U3 - Toxicity prior Beta(0.15, 0.85)	20.9	2	0.6	0	0	
U2 - Toxicity prior Beta(0.3, 0.7)	20.9	2.1	0.6	0	0	
U3 - Toxicity prior Beta(0.3, 0.7)	20.9	2.1	0.6	0	0	
U2 - Toxicity prior Beta(0.5, 0.5)	20.5	2.4	0.7	0.1	0	
U3 - Toxicity prior Beta(0.5, 0.5)	20.6	2.3	0.7	0.1	0	
U2 - Toxicity prior Beta(1, 1)	20.5	2.3	0.7	0.1	0	
U3 - Toxicity prior Beta(1, 1)	20.5	2.3	0.7	0.1	0	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.1	0.7	0.1	0	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.1	0.7	0.1	0	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	0.7	0.1	0	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.1	0.7	0.1	0	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	0.7	0.1	0	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	0.7	0.1	0	0	
U2 - Toxicity prior Beta(1, 1)	0.2	0.7	0.1	0	0	
U3 - Toxicity prior Beta(1, 1)	0.1	0.7	0.1	0	0	
Scenario 22						
DLT rate	0.07	0.27	0.5	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.5	0.5	
Utility 2	0	0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	26.3	55.1	12.4	1.9	0	4.3
U3 - Toxicity prior Beta(0.15, 0.85)	29.5	54.7	10	1.5	0	4.3
U2 - Toxicity prior Beta(0.3, 0.7)	25.1	55.8	12.9	2.1	0	4.1
U3 - Toxicity prior Beta(0.3, 0.7)	31.2	52.3	10.3	2.1	0	4.1
U2 - Toxicity prior Beta(0.5, 0.5)	24	53.1	15.5	3.3	0	4.1
U3 - Toxicity prior Beta(0.5, 0.5)	28.5	51.8	13.1	2.6	0	4
U2 - Toxicity prior Beta(1, 1)	22.3	54	16.3	3.3	0	4.1
U3 - Toxicity prior Beta(1, 1)	28.6	51.5	12.7	3.1	0	4.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	8.7	12.1	6.5	1.7	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	8.7	12.1	6.6	1.7	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	8.6	11.8	6.8	1.8	0.4	
U3 - Toxicity prior Beta(0.3, 0.7)	8.6	11.8	6.7	1.9	0.4	
U2 - Toxicity prior Beta(0.5, 0.5)	8	11.6	6.6	2.7	0.4	
U3 - Toxicity prior Beta(0.5, 0.5)	8	11.7	6.6	2.7	0.5	
U2 - Toxicity prior Beta(1, 1)	8.1	11.6	6.7	2.7	0.4	
U3 - Toxicity prior Beta(1, 1)	8	11.5	6.7	2.8	0.4	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.7	4	1.3	0.5	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.7	4.1	1.3	0.5	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.7	3.9	1.3	0.5	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.7	4	1.3	0.5	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.7	3.6	1.5	0.6	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.7	3.6	1.5	0.6	0.1	
U2 - Toxicity prior Beta(1, 1)	0.7	3.6	1.5	0.6	0.1	
U3 - Toxicity prior Beta(1, 1)	0.7	3.6	1.5	0.6	0.1	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 23						
DLT rate	0.32	0.55	0.8	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	19.6	5.8	0.2	0	0	74.4
U3 - Toxicity prior Beta(0.15, 0.85)	20.2	5.7	0	0	0	74.1
U2 - Toxicity prior Beta(0.3, 0.7)	20.3	6	0.1	0	0	73.6
U3 - Toxicity prior Beta(0.3, 0.7)	21	4.8	0.1	0	0	74.1
U2 - Toxicity prior Beta(0.5, 0.5)	21.9	4.7	0.1	0	0	73.3
U3 - Toxicity prior Beta(0.5, 0.5)	22.6	4.5	0	0	0	72.9
U2 - Toxicity prior Beta(1, 1)	22.1	5	0.2	0	0	72.7
U3 - Toxicity prior Beta(1, 1)	22.6	4.4	0	0	0	73
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	17.2	2.7	0.6	0	0	
U3 - Toxicity prior Beta(0.15, 0.85)	17.2	2.7	0.6	0	0	
U2 - Toxicity prior Beta(0.3, 0.7)	17.2	2.7	0.6	0	0	
U3 - Toxicity prior Beta(0.3, 0.7)	17.2	2.8	0.6	0	0	
U2 - Toxicity prior Beta(0.5, 0.5)	17	3.1	0.7	0.1	0	
U3 - Toxicity prior Beta(0.5, 0.5)	17.1	3.1	0.7	0.1	0	
U2 - Toxicity prior Beta(1, 1)	17	3.1	0.7	0.1	0	
U3 - Toxicity prior Beta(1, 1)	17	3.1	0.7	0.1	0	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.3	1.5	0.2	0	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.3	1.5	0.2	0	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.3	1.5	0.2	0	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.3	1.5	0.2	0	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	1.5	0.2	0	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	1.5	0.2	0	0	
U2 - Toxicity prior Beta(1, 1)	0.4	1.5	0.2	0	0	
U3 - Toxicity prior Beta(1, 1)	0.5	1.5	0.2	0	0	
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	0.9	1.9	4.8	43.8	48.5	0.1
U3 - Toxicity prior Beta(0.15, 0.85)	2.4	2.4	6.5	47.4	41.2	0.1
U2 - Toxicity prior Beta(0.3, 0.7)	0.9	1.1	5.5	44.8	47.7	0
U3 - Toxicity prior Beta(0.3, 0.7)	2.1	2.5	6.4	46.9	42	0.1
U2 - Toxicity prior Beta(0.5, 0.5)	0.9	1.5	4.9	43.3	49.4	0
U3 - Toxicity prior Beta(0.5, 0.5)	2.3	2.2	6	46.4	43.1	0
U2 - Toxicity prior Beta(1, 1)	1.4	1.7	4.8	43.5	48.6	0
U3 - Toxicity prior Beta(1, 1)	2.6	2.3	6.4	46.2	42.5	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.3	3.3	3.8	8.2	11.4	
U3 - Toxicity prior Beta(0.15, 0.85)	3.3	3.3	3.8	8.2	11.4	
U2 - Toxicity prior Beta(0.3, 0.7)	3.3	3.2	3.8	8.1	11.6	
U3 - Toxicity prior Beta(0.3, 0.7)	3.3	3.2	3.8	8.1	11.6	
U2 - Toxicity prior Beta(0.5, 0.5)	3.3	3.2	3.8	8.1	11.6	
U3 - Toxicity prior Beta(0.5, 0.5)	3.3	3.2	3.8	8.1	11.6	
U2 - Toxicity prior Beta(1, 1)	3.3	3.2	3.8	8.1	11.6	
U3 - Toxicity prior Beta(1, 1)	3.3	3.2	3.7	8.1	11.7	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.1	2.5	1.2	3.1	1.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.1	2.5	1.2	3.1	1.2	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	2.5	1.2	3.1	1.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.1	2.5	1.2	3.1	1.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2.5	1.2	3.1	1.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	2.5	1.2	3.2	1.2	
U2 - Toxicity prior Beta(1, 1)	0.1	2.5	1.2	3.1	1.2	
U3 - Toxicity prior Beta(1, 1)	0.1	2.5	1.2	3.1	1.2	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.2	0.25	0.4	0.45	0.5	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	34	28.3	32.3	4.7	0	0.7
U3 - Toxicity prior Beta(0.15, 0.85)	38.7	28.9	27.9	3.8	0	0.7
U2 - Toxicity prior Beta(0.3, 0.7)	34.5	26.6	32.6	5.5	0.1	0.7
U3 - Toxicity prior Beta(0.3, 0.7)	39.4	25.8	29.2	4.9	0	0.7
U2 - Toxicity prior Beta(0.5, 0.5)	34.4	23.8	34.9	6.3	0	0.6
U3 - Toxicity prior Beta(0.5, 0.5)	39.9	26.2	28.5	4.8	0	0.6
U2 - Toxicity prior Beta(1, 1)	33.7	24.7	35.6	5.4	0	0.6
U3 - Toxicity prior Beta(1, 1)	38.9	27.1	28.9	4.5	0	0.6
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9.3	7.8	9.4	2.8	0.5	
U3 - Toxicity prior Beta(0.15, 0.85)	9.4	7.8	9.4	2.8	0.5	
U2 - Toxicity prior Beta(0.3, 0.7)	9.2	7.6	9.3	3.2	0.6	
U3 - Toxicity prior Beta(0.3, 0.7)	9.2	7.5	9.5	3.1	0.6	
U2 - Toxicity prior Beta(0.5, 0.5)	8.8	7.1	9.5	3.8	0.7	
U3 - Toxicity prior Beta(0.5, 0.5)	8.8	7	9.5	3.9	0.7	
U2 - Toxicity prior Beta(1, 1)	8.8	7.1	9.4	3.9	0.7	
U3 - Toxicity prior Beta(1, 1)	8.8	7.1	9.5	3.9	0.7	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.7	3.1	1.5	0.8	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.7	3.1	1.5	0.9	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.7	3	1.6	0.9	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.7	3	1.6	0.9	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.7	2.9	1.7	0.9	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.7	2.9	1.7	0.9	0.1	
U2 - Toxicity prior Beta(1, 1)	0.7	2.8	1.7	0.9	0.1	
U3 - Toxicity prior Beta(1, 1)	0.7	2.9	1.7	0.9	0.1	
Scenario 26						
DLT rate	0.15	0.22	0.3	0.62	0.8	
Efficacy rate	0.1	0.2	0.35	0.45	0.55	
Utility 2	-0.1	0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	29	26.5	28.5	5.8	0.1	10.1
U3 - Toxicity prior Beta(0.15, 0.85)	34.3	26.1	24.2	5.2	0	10.2
U2 - Toxicity prior Beta(0.3, 0.7)	29.1	25.7	28.4	6.6	0.1	10.1
U3 - Toxicity prior Beta(0.3, 0.7)	32.6	26.7	24.9	5.6	0.1	10.1
U2 - Toxicity prior Beta(0.5, 0.5)	27.1	23.7	31.1	8.1	0.1	9.9
U3 - Toxicity prior Beta(0.5, 0.5)	32.2	23.6	26.9	7.3	0	10
U2 - Toxicity prior Beta(1, 1)	26.9	24.2	30.7	7.9	0.2	10.1
U3 - Toxicity prior Beta(1, 1)	31.3	23.5	27.3	7.7	0.1	10.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	11	6.5	8	2.8	0.5	
U3 - Toxicity prior Beta(0.15, 0.85)	11	6.5	7.9	2.9	0.5	
U2 - Toxicity prior Beta(0.3, 0.7)	10.8	6.3	8	3.2	0.6	
U3 - Toxicity prior Beta(0.3, 0.7)	10.7	6.2	8.1	3.2	0.6	
U2 - Toxicity prior Beta(0.5, 0.5)	10.3	6	8.3	3.8	0.7	
U3 - Toxicity prior Beta(0.5, 0.5)	10.3	5.8	8.3	3.8	0.7	
U2 - Toxicity prior Beta(1, 1)	10.2	5.9	8.4	3.8	0.7	
U3 - Toxicity prior Beta(1, 1)	10.3	5.8	8.4	3.8	0.7	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	2.9	1.3	0.9	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	3	1.3	0.9	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	2.9	1.3	0.9	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	2.9	1.3	0.9	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	2.7	1.4	1	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	2.7	1.4	1	0.1	
U2 - Toxicity prior Beta(1, 1)	0.5	2.7	1.4	0.9	0.1	
U3 - Toxicity prior Beta(1, 1)	0.5	2.7	1.4	0.9	0.1	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
Utility 2	0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	21.2	21.2	41.9	9.2	1	5.5
U3 - Toxicity prior Beta(0.15, 0.85)	25.8	22.9	37.6	7.7	0.6	5.4
U2 - Toxicity prior Beta(0.3, 0.7)	20.5	20.4	42.9	9.9	1	5.3
U3 - Toxicity prior Beta(0.3, 0.7)	25.5	21.2	38	9.2	0.8	5.3
U2 - Toxicity prior Beta(0.5, 0.5)	20.3	17.5	43.8	11.6	1.4	5.4
U3 - Toxicity prior Beta(0.5, 0.5)	23.4	20.1	39.5	10.6	1.1	5.3
U2 - Toxicity prior Beta(1, 1)	20.2	16.8	44.1	12.1	1.4	5.4
U3 - Toxicity prior Beta(1, 1)	24.6	18.6	39.5	10.7	1.2	5.4
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	8.3	6.3	9.4	4	1.4	
U3 - Toxicity prior Beta(0.15, 0.85)	8.2	6.4	9.4	4	1.4	
U2 - Toxicity prior Beta(0.3, 0.7)	8.1	6	9.4	4.3	1.6	
U3 - Toxicity prior Beta(0.3, 0.7)	8.1	6	9.3	4.3	1.6	
U2 - Toxicity prior Beta(0.5, 0.5)	7.7	5.5	9.3	5.2	1.7	
U3 - Toxicity prior Beta(0.5, 0.5)	7.7	5.6	9.3	5	1.8	
U2 - Toxicity prior Beta(1, 1)	7.7	5.5	9.3	5.1	1.7	
U3 - Toxicity prior Beta(1, 1)	7.7	5.6	9.2	5.2	1.8	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	3	1.6	1.1	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	2.9	1.6	1.1	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	2.9	1.6	1.1	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	2.9	1.6	1.1	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	2.7	1.8	1.2	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	2.7	1.8	1.2	0.3	
U2 - Toxicity prior Beta(1, 1)	0.4	2.7	1.8	1.2	0.3	
U3 - Toxicity prior Beta(1, 1)	0.4	2.7	1.7	1.2	0.3	
Scenario 28						
DLT rate	0.2	0.3	0.5	0.6	0.77	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	21.6	37.6	6.8	0.3	0	33.7
U3 - Toxicity prior Beta(0.15, 0.85)	21.8	38.8	5.7	0.3	0	33.4
U2 - Toxicity prior Beta(0.3, 0.7)	20.2	39.8	6.5	0.6	0	32.9
U3 - Toxicity prior Beta(0.3, 0.7)	21.3	39.7	5.7	0.4	0	32.9
U2 - Toxicity prior Beta(0.5, 0.5)	21.8	38.6	7.1	0.9	0	31.6
U3 - Toxicity prior Beta(0.5, 0.5)	22.9	38.9	5.7	0.9	0	31.6
U2 - Toxicity prior Beta(1, 1)	20.8	38.3	7.9	1.1	0	31.9
U3 - Toxicity prior Beta(1, 1)	22.8	38.7	5.8	0.8	0	31.9
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	12	8.4	4.3	0.9	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	12.1	8.4	4.3	0.9	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	12	8.2	4.5	1	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	11.9	8.2	4.5	1	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	11.2	8.8	4.5	1.4	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	11.2	8.8	4.5	1.4	0.2	
U2 - Toxicity prior Beta(1, 1)	11.2	8.7	4.5	1.5	0.2	
U3 - Toxicity prior Beta(1, 1)	11.2	8.7	4.6	1.4	0.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.3	2.7	0.9	0.3	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.3	2.7	0.8	0.3	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.3	2.6	0.9	0.3	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.3	2.7	0.9	0.3	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	2.5	1	0.3	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	2.5	1	0.3	0	
U2 - Toxicity prior Beta(1, 1)	0.3	2.4	1	0.3	0	
U3 - Toxicity prior Beta(1, 1)	0.3	2.5	1	0.3	0	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 29						
DLT rate	0.19	0.3	0.56	0.74	0.86	
Efficacy rate	0.4	0.4	0.4	0.4	0.4	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	69.7	23.3	4.1	0.3	0	2.6
U3 - Toxicity prior Beta(0.15, 0.85)	72.4	21.5	3.2	0.3	0	2.6
U2 - Toxicity prior Beta(0.3, 0.7)	69.1	24.1	3.8	0.4	0	2.6
U3 - Toxicity prior Beta(0.3, 0.7)	72.2	22.2	3	0.1	0	2.5
U2 - Toxicity prior Beta(0.5, 0.5)	69	23.7	4.7	0.3	0	2.3
U3 - Toxicity prior Beta(0.5, 0.5)	71.7	22.3	3.5	0.2	0	2.3
U2 - Toxicity prior Beta(1, 1)	68.5	24.6	4.5	0.1	0	2.3
U3 - Toxicity prior Beta(1, 1)	70.6	23	3.7	0.2	0	2.5
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	18.2	7.1	3.6	0.5	0	
U3 - Toxicity prior Beta(0.15, 0.85)	18.3	7.1	3.5	0.6	0	
U2 - Toxicity prior Beta(0.3, 0.7)	18	7	3.8	0.6	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	18.1	7	3.7	0.7	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	17.5	7.3	3.8	0.8	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	17.4	7.4	3.8	0.9	0.1	
U2 - Toxicity prior Beta(1, 1)	17.4	7.4	3.8	0.9	0.1	
U3 - Toxicity prior Beta(1, 1)	17.4	7.4	3.8	0.9	0.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.6	2.1	0.8	0.2	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.7	2.1	0.8	0.2	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.6	2	0.8	0.2	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.6	2	0.8	0.2	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.7	2	0.8	0.2	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.7	2	0.8	0.2	0	
U2 - Toxicity prior Beta(1, 1)	0.7	2	0.8	0.2	0	
U3 - Toxicity prior Beta(1, 1)	0.7	1.9	0.8	0.2	0	
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.2	0.4	0.5	
Utility 2	0.1	0.1	0	0.1	0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	21	13.2	15.7	37	12.3	0.8
U3 - Toxicity prior Beta(0.15, 0.85)	27.8	14.5	13.8	32.9	10.2	0.8
U2 - Toxicity prior Beta(0.3, 0.7)	19.9	12.8	15.3	38.3	12.9	0.8
U3 - Toxicity prior Beta(0.3, 0.7)	27.6	14.2	12.9	34.2	10.3	0.8
U2 - Toxicity prior Beta(0.5, 0.5)	17.6	12.1	11.8	41.5	16.2	0.8
U3 - Toxicity prior Beta(0.5, 0.5)	26.4	13.7	10.9	36	12.2	0.8
U2 - Toxicity prior Beta(1, 1)	16.6	12.5	12.4	41	16.7	0.8
U3 - Toxicity prior Beta(1, 1)	25.3	14.7	10.3	36.1	12.8	0.8
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	5.2	4.5	5.7	8.1	6.5	
U3 - Toxicity prior Beta(0.15, 0.85)	5.2	4.5	5.7	8.1	6.5	
U2 - Toxicity prior Beta(0.3, 0.7)	5.1	4.4	5.5	8.1	6.8	
U3 - Toxicity prior Beta(0.3, 0.7)	5.1	4.4	5.4	8.1	6.8	
U2 - Toxicity prior Beta(0.5, 0.5)	5	4.2	5.2	8.6	7	
U3 - Toxicity prior Beta(0.5, 0.5)	5	4.2	5.2	8.6	7	
U2 - Toxicity prior Beta(1, 1)	4.9	4.1	5.2	8.6	7.1	
U3 - Toxicity prior Beta(1, 1)	4.9	4.2	5.2	8.5	7.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	2.7	1.5	2.3	0.9	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	2.8	1.5	2.3	0.9	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	2.7	1.5	2.3	1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	2.7	1.5	2.3	1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	2.7	1.5	2.3	1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	2.7	1.5	2.3	1	
U2 - Toxicity prior Beta(1, 1)	0.3	2.7	1.5	2.3	1	
U3 - Toxicity prior Beta(1, 1)	0.3	2.7	1.5	2.3	1	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 31						
DLT rate	0.04	0.06	0.1	0.2	0.6	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	7.6	35	25.6	25.2	6.5	0.1
U3 - Toxicity prior Beta(0.15, 0.85)	10.3	41.9	24.3	19.5	3.9	0.1
U2 - Toxicity prior Beta(0.3, 0.7)	7	35.7	25.1	25.1	7	0.1
U3 - Toxicity prior Beta(0.3, 0.7)	9.1	42.6	22.6	21.2	4.4	0.1
U2 - Toxicity prior Beta(0.5, 0.5)	7.6	32.9	25.2	25.5	8.7	0.1
U3 - Toxicity prior Beta(0.5, 0.5)	10.2	41.5	21.4	21.5	5.3	0.1
U2 - Toxicity prior Beta(1, 1)	7	34.3	24.6	24.3	9.7	0.1
U3 - Toxicity prior Beta(1, 1)	9.7	41.1	22.4	21.1	5.6	0.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.1	5.7	6.4	7.7	6.1	
U3 - Toxicity prior Beta(0.15, 0.85)	4.1	5.7	6.4	7.6	6	
U2 - Toxicity prior Beta(0.3, 0.7)	4.1	5.7	6.2	7.6	6.3	
U3 - Toxicity prior Beta(0.3, 0.7)	4.1	5.7	6.2	7.7	6.3	
U2 - Toxicity prior Beta(0.5, 0.5)	4.1	5.6	6.2	7.6	6.5	
U3 - Toxicity prior Beta(0.5, 0.5)	4.1	5.7	6.2	7.6	6.5	
U2 - Toxicity prior Beta(1, 1)	4.1	5.7	6.2	7.6	6.5	
U3 - Toxicity prior Beta(1, 1)	4.1	5.6	6.2	7.6	6.5	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.1	2.8	1.4	1.7	1.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.1	2.8	1.4	1.7	1.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	2.8	1.4	1.7	1.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.1	2.8	1.4	1.8	1.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.7	1.4	1.7	1.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.7	1.4	1.7	1.1	
U2 - Toxicity prior Beta(1, 1)	0.1	2.7	1.4	1.7	1.2	
U3 - Toxicity prior Beta(1, 1)	0.1	2.7	1.4	1.7	1.2	
Scenario 32						
DLT rate	0.02	0.05	0.11	0.3	0.36	
Efficacy rate	0.08	0.15	0.3	0.3	0.3	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	9.5	13.2	35	27.3	14.4	0.6
U3 - Toxicity prior Beta(0.15, 0.85)	15.4	16.5	34	24.1	9.4	0.6
U2 - Toxicity prior Beta(0.3, 0.7)	9.1	12.8	34.2	28.4	14.9	0.6
U3 - Toxicity prior Beta(0.3, 0.7)	14.5	17.2	34.4	24.2	9.1	0.6
U2 - Toxicity prior Beta(0.5, 0.5)	8.3	13.5	34.4	27.8	15.4	0.6
U3 - Toxicity prior Beta(0.5, 0.5)	14.7	16.3	34.7	23.6	10.1	0.6
U2 - Toxicity prior Beta(1, 1)	9.3	12.2	33.7	28.1	16.1	0.6
U3 - Toxicity prior Beta(1, 1)	14.6	16.8	33.9	24.4	9.7	0.6
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4	4.3	7.6	7	7	
U3 - Toxicity prior Beta(0.15, 0.85)	4	4.3	7.6	7	7	
U2 - Toxicity prior Beta(0.3, 0.7)	4	4.3	7.4	7	7.2	
U3 - Toxicity prior Beta(0.3, 0.7)	4	4.3	7.5	7	7.2	
U2 - Toxicity prior Beta(0.5, 0.5)	4	4.2	7.4	7.1	7.3	
U3 - Toxicity prior Beta(0.5, 0.5)	4	4.2	7.3	7.1	7.3	
U2 - Toxicity prior Beta(1, 1)	4	4.2	7.3	7.2	7.3	
U3 - Toxicity prior Beta(1, 1)	4	4.2	7.3	7.1	7.3	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.3	2.9	2	2.6	1.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.3	2.9	2.1	2.6	1.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.3	2.8	2.1	2.5	1.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.3	2.9	2	2.5	1.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	2.8	2.1	2.5	1.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	2.8	2.1	2.5	1.2	
U2 - Toxicity prior Beta(1, 1)	0.3	2.8	2.1	2.5	1.2	
U3 - Toxicity prior Beta(1, 1)	0.3	2.8	2.1	2.5	1.2	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.2	0.4	0.5	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	7.3	5.6	12.7	40	34.3	0.1
U3 - Toxicity prior Beta(0.15, 0.85)	12.6	6.6	12.4	40.6	27.8	0
U2 - Toxicity prior Beta(0.3, 0.7)	7.9	4.9	12.8	38.5	35.8	0.1
U3 - Toxicity prior Beta(0.3, 0.7)	13.1	6.6	12.4	38.3	29.5	0.1
U2 - Toxicity prior Beta(0.5, 0.5)	7.2	4.1	11.6	37.9	39.2	0
U3 - Toxicity prior Beta(0.5, 0.5)	11.8	6.5	11.9	37.5	32.3	0
U2 - Toxicity prior Beta(1, 1)	6.1	4.3	11.3	38.5	39.8	0
U3 - Toxicity prior Beta(1, 1)	11.3	5.8	11.5	39.4	32	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.5	3.6	5.2	8	9.7	
U3 - Toxicity prior Beta(0.15, 0.85)	3.5	3.7	5.2	8	9.7	
U2 - Toxicity prior Beta(0.3, 0.7)	3.5	3.6	5	7.9	10.1	
U3 - Toxicity prior Beta(0.3, 0.7)	3.5	3.6	5	7.9	10	
U2 - Toxicity prior Beta(0.5, 0.5)	3.4	3.5	4.7	8.2	10.2	
U3 - Toxicity prior Beta(0.5, 0.5)	3.4	3.5	4.7	8.2	10.1	
U2 - Toxicity prior Beta(1, 1)	3.4	3.5	4.7	8.2	10.2	
U3 - Toxicity prior Beta(1, 1)	3.4	3.5	4.7	8.2	10.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	2.6	1.4	2.8	1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	2.6	1.4	2.8	1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.6	1.4	2.7	1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	2.6	1.4	2.7	1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2.6	1.4	2.7	1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	2.6	1.4	2.7	1	
U2 - Toxicity prior Beta(1, 1)	0.1	2.6	1.4	2.7	1	
U3 - Toxicity prior Beta(1, 1)	0.1	2.6	1.4	2.7	1	
Scenario 34						
DLT rate	0.05	0.13	0.3	0.36	0.42	
Efficacy rate	0.05	0.1	0.3	0.32	0.35	
Utility 2	0	0	0	0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	13.4	12.2	47.8	18.4	6.2	2
U3 - Toxicity prior Beta(0.15, 0.85)	19.7	13.9	44.9	16.4	3.2	1.9
U2 - Toxicity prior Beta(0.3, 0.7)	12.9	12	46.3	20.3	6.6	1.9
U3 - Toxicity prior Beta(0.3, 0.7)	20.3	13.2	42.5	17.8	4.2	2
U2 - Toxicity prior Beta(0.5, 0.5)	11.3	11	44.8	23.1	7.9	1.9
U3 - Toxicity prior Beta(0.5, 0.5)	18.5	13	41.3	19.9	5.4	1.9
U2 - Toxicity prior Beta(1, 1)	10.8	11.8	43.7	23.1	8.7	1.9
U3 - Toxicity prior Beta(1, 1)	17.6	13.7	41.5	20.5	4.8	1.9
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	5.3	5.6	10	5.3	3.5	
U3 - Toxicity prior Beta(0.15, 0.85)	5.4	5.7	9.9	5.3	3.5	
U2 - Toxicity prior Beta(0.3, 0.7)	5.1	5.4	9.8	5.6	3.9	
U3 - Toxicity prior Beta(0.3, 0.7)	5.2	5.4	9.8	5.5	3.9	
U2 - Toxicity prior Beta(0.5, 0.5)	4.9	5.1	9.2	6.3	4.3	
U3 - Toxicity prior Beta(0.5, 0.5)	4.9	5	9.2	6.3	4.3	
U2 - Toxicity prior Beta(1, 1)	4.9	5	9.1	6.4	4.3	
U3 - Toxicity prior Beta(1, 1)	4.9	5	9.2	6.4	4.3	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	3.3	1.9	1.7	0.6	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	3.3	2	1.7	0.6	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	3.2	1.9	1.7	0.6	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	3.2	1.9	1.7	0.6	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	3	2.1	1.8	0.7	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	3	2	1.7	0.7	
U2 - Toxicity prior Beta(1, 1)	0.4	3	2.1	1.8	0.7	
U3 - Toxicity prior Beta(1, 1)	0.4	3	2.1	1.7	0.7	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 35						
DLT rate	0.03	0.09	0.3	0.4	0.45	
Efficacy rate	0.2	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	12.8	39.8	31.2	11.9	4.2	0.1
U3 - Toxicity prior Beta(0.15, 0.85)	16.4	44.1	26.4	10.4	2.6	0.1
U2 - Toxicity prior Beta(0.3, 0.7)	11.2	39.4	31.4	12.9	5	0.1
U3 - Toxicity prior Beta(0.3, 0.7)	15.5	44.6	25.5	11.6	2.7	0.1
U2 - Toxicity prior Beta(0.5, 0.5)	10.7	39.3	29.3	15.3	5.3	0.1
U3 - Toxicity prior Beta(0.5, 0.5)	14.9	44	24.3	13.5	3.2	0.1
U2 - Toxicity prior Beta(1, 1)	10.8	40.4	28.5	14.6	5.6	0.1
U3 - Toxicity prior Beta(1, 1)	14.3	43.8	25.5	13.3	3	0.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.7	8.2	8.9	5	3.2	
U3 - Toxicity prior Beta(0.15, 0.85)	4.7	8.3	8.9	5	3.2	
U2 - Toxicity prior Beta(0.3, 0.7)	4.6	7.9	8.7	5.2	3.5	
U3 - Toxicity prior Beta(0.3, 0.7)	4.7	7.9	8.7	5.2	3.5	
U2 - Toxicity prior Beta(0.5, 0.5)	4.6	7.4	8.4	5.9	3.7	
U3 - Toxicity prior Beta(0.5, 0.5)	4.5	7.4	8.4	5.9	3.7	
U2 - Toxicity prior Beta(1, 1)	4.6	7.4	8.3	5.9	3.7	
U3 - Toxicity prior Beta(1, 1)	4.6	7.4	8.4	6	3.7	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.3	3.1	1.6	1.4	0.5	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	3.1	1.6	1.4	0.5	
U2 - Toxicity prior Beta(0.3, 0.7)	0.3	3.1	1.6	1.4	0.5	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	3.1	1.6	1.4	0.5	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	3	1.7	1.4	0.6	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	3	1.7	1.4	0.6	
U2 - Toxicity prior Beta(1, 1)	0.3	3	1.7	1.5	0.5	
U3 - Toxicity prior Beta(1, 1)	0.3	3	1.7	1.5	0.5	
Scenario 36						
DLT rate	0.13	0.29	0.35	0.4	0.43	
Efficacy rate	0.05	0.2	0.22	0.3	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	20.4	40.2	15.9	6.3	2.3	14.9
U3 - Toxicity prior Beta(0.15, 0.85)	24.5	42.5	11.6	4.9	1.4	15.1
U2 - Toxicity prior Beta(0.3, 0.7)	19.3	39.4	16.4	7.8	2.3	14.8
U3 - Toxicity prior Beta(0.3, 0.7)	26.1	38.7	13.3	5.6	1.5	14.8
U2 - Toxicity prior Beta(0.5, 0.5)	18.3	37.7	17.2	9.3	3.4	14.1
U3 - Toxicity prior Beta(0.5, 0.5)	24	37.4	14.5	8.1	1.8	14.2
U2 - Toxicity prior Beta(1, 1)	18.5	35.6	18.4	10.2	3	14.3
U3 - Toxicity prior Beta(1, 1)	24.7	37.1	14.3	7.9	1.8	14.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9.6	8.2	5.8	2.8	1.4	
U3 - Toxicity prior Beta(0.15, 0.85)	9.6	8.2	5.7	2.8	1.4	
U2 - Toxicity prior Beta(0.3, 0.7)	9.5	7.9	5.9	3	1.6	
U3 - Toxicity prior Beta(0.3, 0.7)	9.4	8	5.9	2.9	1.6	
U2 - Toxicity prior Beta(0.5, 0.5)	8.7	8.1	5.9	3.5	1.9	
U3 - Toxicity prior Beta(0.5, 0.5)	8.7	8.1	5.8	3.6	1.9	
U2 - Toxicity prior Beta(1, 1)	8.8	8.1	5.8	3.6	1.9	
U3 - Toxicity prior Beta(1, 1)	8.7	8	5.9	3.6	1.9	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	3.5	1.2	0.9	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	3.6	1.2	0.9	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	3.4	1.2	1	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	3.4	1.2	0.9	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	3.2	1.4	1	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	3.2	1.4	1	0.3	
U2 - Toxicity prior Beta(1, 1)	0.5	3.2	1.4	1	0.3	
U3 - Toxicity prior Beta(1, 1)	0.5	3.2	1.4	1	0.3	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 37						
DLT rate	0.08	0.3	0.4	0.45	0.5	
Efficacy rate	0.05	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	26.2	49.1	14.1	3.1	0.4	7.1
U3 - Toxicity prior Beta(0.15, 0.85)	32.5	47.7	10	2.4	0.3	7.1
U2 - Toxicity prior Beta(0.3, 0.7)	24.6	50.4	13.9	3.9	0.6	6.6
U3 - Toxicity prior Beta(0.3, 0.7)	32.5	47.4	9.8	2.9	0.4	7
U2 - Toxicity prior Beta(0.5, 0.5)	22.5	47.7	17.1	4.9	1	6.8
U3 - Toxicity prior Beta(0.5, 0.5)	31	45.9	12.4	3.4	0.7	6.6
U2 - Toxicity prior Beta(1, 1)	23	46.7	17.4	5	1.1	6.8
U3 - Toxicity prior Beta(1, 1)	31.1	45.7	12.5	3.2	0.8	6.7
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	8.7	10.7	6.1	2.4	0.9	
U3 - Toxicity prior Beta(0.15, 0.85)	8.6	10.7	6.1	2.4	0.9	
U2 - Toxicity prior Beta(0.3, 0.7)	8.6	10.2	6.3	2.6	1.1	
U3 - Toxicity prior Beta(0.3, 0.7)	8.6	10.3	6.3	2.5	1.1	
U2 - Toxicity prior Beta(0.5, 0.5)	7.9	10.4	6.2	3.1	1.3	
U3 - Toxicity prior Beta(0.5, 0.5)	7.8	10.4	6.3	3.1	1.3	
U2 - Toxicity prior Beta(1, 1)	7.9	10.4	6.3	3.1	1.2	
U3 - Toxicity prior Beta(1, 1)	7.8	10.4	6.3	3.1	1.3	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.6	4.1	1.3	0.9	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.6	4.1	1.3	0.9	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	0.6	4	1.3	0.9	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.6	4	1.3	0.9	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.6	3.8	1.5	1	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	0.6	3.7	1.5	1	0.3	
U2 - Toxicity prior Beta(1, 1)	0.6	3.7	1.5	1	0.3	
U3 - Toxicity prior Beta(1, 1)	0.6	3.7	1.6	1	0.3	